

ON THE MOMENTS OF THE TRACE OF NONCENTRAL WISHART MATRICES AND SUBMATRICES

D.G. Nel*

University of the Orange Free State

1. Introduction

James (1964) derived an expression for the h -th moment of the trace of a central Wishart matrix in terms of zonal polynomials. Khatri and Pillai (1967) calculated the first four moments of $\text{tr}A$ in terms of the elementary symmetric functions of the covariance matrix Σ . Nel (1971) derived an expression for the h -th moment of the trace of a noncentral Wishart matrix with covariance matrix $\Sigma = \sigma^2 I_p$. Herz (1955) derived the moment generating function of a noncentral Wishart matrix.

In this paper the moment generating function of a noncentral Wishart matrix, is used to derive the first six moments of the trace of A as well as the trace of submatrices of A . A conjecture concerning the h -th moment of $\text{tr}A$ is stated.

2. Some useful results on the differentiation of the elementary symmetrical functions.

Denote the k -th elementary symmetric function of a positive definite ($p \times p$) matrix A by $\text{tr}_k A$.

Lemma 1

$$2.1 \quad |A| \text{tr}_{p-j} A^{-1} = \text{tr}_j A.$$

Proof

Let $\lambda_1 > \lambda_2 > \dots > \lambda_p > 0$ be the different latent roots of A . Then $|A| = \prod_{i=1}^p \lambda_i$.

The corresponding latent roots of A^{-1} are: $0 < \theta_1 < \theta_2 < \dots < \theta_p$, where $\theta_i = 1/\lambda_i$ ($i=1, \dots, p$).

$$\begin{aligned} \text{Since } \text{tr}_j A^{-1} &= \sum_{1 \leq i_1 < \dots < i_j \leq p} \theta_{i_1} \dots \theta_{i_j} \\ \text{it follows that: } \text{tr}_{p-j} A^{-1} &= \sum_{1 \leq i_1 < \dots < i_{p-j} \leq p} \theta_{i_1} \dots \theta_{i_{p-j}} \\ &= \sum_{1 \leq i_1 < \dots < i_{p-j} \leq p} 1/(\lambda_{i_1} \dots \lambda_{i_{p-j}}) \end{aligned}$$

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$$\begin{aligned}
|A| \operatorname{tr}_{p-j} A^{-1} &= (\prod_{i=1}^p \lambda_i) \sum_{1 \leq i_1 < \dots < i_{p-j} \leq p} 1/(\lambda_{i_1} \dots \lambda_{i_{p-j}}) \\
&= \sum_{1 \leq i_1 < \dots < i_j \leq p} \lambda_{i_1} \dots \lambda_{i_j} \\
&= \operatorname{tr}_j A
\end{aligned}$$

since the number of terms in $\operatorname{tr}_{p-j} A^{-1}$ equals $\binom{p}{p-j} = \binom{p}{j}$, which is the number of terms in $\operatorname{tr}_j A$.

Lemma 2

If λ is any scalar constant and β a scalar variable, then:

$$2.2 \quad \frac{d}{d\beta} (\operatorname{tr}_j(A + \lambda\beta I)) = \lambda(p-j+1) \operatorname{tr}_{j-1}(A + \lambda\beta I)$$

Proof

Let $\theta_1 > \theta_2 > \dots > \theta_p$ be the different latent roots of $A + \lambda\beta I$. Then

$$\operatorname{tr}_j(A + \lambda\beta I) = \sum_{1 \leq i_1 < \dots < i_j \leq p} \theta_{i_1} \dots \theta_{i_j}$$

where $\theta_{i_\alpha} = \nu_{i_\alpha} + \lambda\beta$ and ν_{i_α} ($\alpha = 1, \dots, p$) the latent roots of A .

Now:

$$2.3 \quad \frac{d}{d\beta} (\operatorname{tr}_j(A + \lambda\beta I)) = \lambda \sum_{1 \leq i_1 < \dots < i_j \leq p} \{(\theta_{i_2} \dots \theta_{i_j}) + \dots + (\theta_{i_1} \dots \theta_{i_{j-1}})\}$$

$$2.4 \quad = \lambda(p-j+1) \sum_{1 \leq i_1 < \dots < i_{j-1} \leq p} \theta_{i_1} \dots \theta_{i_{j-1}}$$

$$= \lambda(p-j+1) \operatorname{tr}_{j-1}(A + \lambda\beta I)$$

Corollary 1

$$2.5 \quad \frac{d^h}{d\beta^h} (|A + \lambda\beta I|) = \lambda^h h! \operatorname{tr}_{p-h}(A + \lambda\beta I)$$

Proof

The result follows immediately by induction when using the fact that $|A + \lambda\beta I| = \operatorname{tr}_p(A + \lambda\beta I)$ and applying Lemma 2 successively.

Corollary 2

$$\frac{d}{d\beta} (\operatorname{tr}_j(A + \lambda\beta I)^{-1}) = \lambda \{ (j+1) \operatorname{tr}_{j+1}(A + \lambda\beta I)^{-1} - \operatorname{tr}_1(A + \lambda\beta I)^{-1} \operatorname{tr}_j(A + \lambda\beta I)^{-1} \}$$

Proof

From Lemma 1 it follows that:

$$\text{tr}_j(A + \lambda\beta I)^{-1} = |A + \lambda\beta I|^{-1} \text{tr}_{p-j}(A + \lambda\beta I)$$

Thus:

$$\begin{aligned} \frac{d}{d\beta} (\text{tr}_j(A + \lambda\beta I)^{-1}) &= |A + \lambda\beta I|^{-1} \lambda(j+1) \text{tr}_{p-j-1}(A + \lambda\beta I) \\ &\quad - |A + \lambda\beta I|^{-2} \lambda \text{tr}_{p-1}(A + \lambda\beta I) \text{tr}_{p-j}(A + \lambda\beta I), \text{ from Lemma 2.} \end{aligned}$$

Apply Lemma 1 and the Corollary is proved.

Lemma 3 (Dwyer 1967)

If $Y = [y_{ij}(x)]$ is a matrix function of the scalar x , then:

$$2.6 \quad \frac{\partial Y}{\partial x} = \left[\frac{\partial y_{ij}}{\partial x} \right]$$

$$2.7 \quad \frac{\partial Y^{-1}}{\partial x} = -Y^{-1} \cdot \frac{\partial Y}{\partial x} \cdot Y^{-1}$$

$$2.8 \quad \frac{\partial(\text{tr } Y)}{\partial x} = \text{tr} \left[\frac{\partial Y}{\partial x} \right]$$

3. The Calculation of the Moments

Let $A(p \times p)$ be distributed according to a noncentral Wishart distribution with n degrees of freedom, covariance matrix Σ and noncentrality parameter Ω . Using the density function of A (Constantine 1963), the moment generating function of A is given by:

$$3.1 \quad M_A(\theta) = E(\text{etr } \theta A) \quad (\theta \text{ a } (p \times p) \text{ matrix})$$

$$\begin{aligned} &= \text{etr}(-\frac{1}{2}\Omega) (1/\Gamma_p(\frac{1}{2}n)) |2\Sigma|^{-\frac{1}{2}n} \sum_{k=0}^{\infty} \Sigma_K^{-1} / ((\frac{1}{2}n)_K k!) \\ &\cdot \int_{A \geq 0} \text{etr}(-\frac{1}{2}(\Sigma^{-1} - 2\theta)A) |A|^{\frac{1}{2}(n-p-1)} C_K(\frac{1}{2}\Omega \Sigma^{-1} A) dA \end{aligned}$$

$$3.2 \quad = \text{etr}(-\frac{1}{2}\Omega) |2\Sigma|^{-\frac{1}{2}n} |2(\Sigma^{-1} - 2\theta)^{-1}|^{\frac{1}{2}n} \text{etr}(\frac{1}{2}\Omega \Sigma^{-1} (\Sigma^{-1} - 2\theta)^{-1})$$

when using Constantine's integral (1963).

thus

$$3.3 \quad M_A(\theta) = |I - 2\theta \Sigma|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega + \frac{1}{2}\Omega(I - 2\theta \Sigma)^{-1})$$

$$3.4 \quad = |I - 2\theta \Sigma|^{-\frac{1}{2}n} \text{etr}(\Omega \theta \Sigma (I - 2\theta \Sigma)^{-1})$$

Hence the moment generating function of $\text{tr}_1 A$ is:

$$3.5 \quad M(\beta) = M_{\text{tr}_1 A}(\beta), \quad \beta \text{ a scalar}$$

$$= |I - 2\beta \Sigma|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega + \frac{1}{2}\Omega(I - 2\beta \Sigma)^{-1})$$

Using the results of section 2 it follows that:

$$\frac{d}{d\beta} M(\beta) = M(\beta) X_1(\beta),$$

$$\text{where } X_1(\beta) = n \text{tr}_1(\Sigma^{-1} - 2\beta I) + \text{tr}_1(\Omega(I - 2\beta \Sigma)^{-1} \Sigma(I - 2\beta \Sigma)^{-1})$$

If we denote $X_1(\beta)|_{\beta=0}$ by X_1 , then:

$$3.6 \quad E(\text{tr } A) = \frac{d}{d\beta} (M(\beta))|_{\beta=0} = X_1 = n \text{tr}_1 \Sigma + \text{tr}_1 \Sigma \Omega.$$

This result was also derived by De Waal (1971).

To calculate $E(\text{tr}^2 A)$ the derivative of $X_1(\beta)$ is required

$$\begin{aligned} \frac{d}{d\beta} (X_1(\beta)) &= 2 \{ n \text{tr}_1^2 (\Sigma^{-1} - 2\beta I)^{-1} \\ &\quad - 2n \text{tr}_2(\Sigma^{-1} - 2\beta I)^{-1} + 2\text{tr}_1 \Omega(I - 2\beta \Sigma) (\Sigma(I - 2\beta \Sigma))^{-2} \} \\ &= 2X_2(\beta), \text{ by denoting the right hand factor by } X_2(\beta) \end{aligned}$$

$$\text{Then: } \frac{d^2}{d\beta^2} (M(\beta)) = M(\beta) \{ X_1^2(\beta) + 2X_2(\beta) \}$$

Thus

$$3.7 \quad E(\text{tr}^2 A) = \frac{d^2}{d\beta^2} (M(\beta))|_{\beta=0} = X_1^2 + 2X_2$$

$$\text{where } X_2 = n \operatorname{tr}_1^2 \Sigma - 2n \operatorname{tr}_2 \Sigma + 2 \operatorname{tr} \Sigma^2 \Omega = n \operatorname{tr}_1 \Sigma^2 + 2 \operatorname{tr}_1 \Sigma^2 \Omega$$

Differentiating $X_2(\beta)$ we get:

$$\frac{d}{d\beta} (X_2(\beta)) = 4X_3(\beta)$$

$$\begin{aligned} \text{where: } X_3(\beta) = & n \operatorname{tr}_1^3 (\Sigma^{-1} - 2\beta I)^{-1} - 3n \operatorname{tr}_1 (\Sigma^{-1} - 2\beta I)^{-1} \operatorname{tr}_2 (\Sigma^{-1} - 2\beta I)^{-1} + \\ & 3n \operatorname{tr}_3 (\Sigma^{-1} - 2\beta I)^{-1} + 3 \operatorname{tr}_1 \Omega (I - 2\beta \Sigma)^{-1} (\Sigma (I - 2\beta \Sigma)^{-1})^3 \end{aligned}$$

Then:

$$3.8 \quad E(\operatorname{tr}^3 A) = \frac{d^3}{d\beta^3} (M(\beta))|_{\beta=0} = X_1^3 + 6 X_1 X_2 + 8 X_3$$

$$\text{where: } X_3 = n \operatorname{tr}_1^3 \Sigma - 3n \operatorname{tr}_1 \Sigma \operatorname{tr}_2 \Sigma + 3n \operatorname{tr}_3 \Sigma + 3 \operatorname{tr}_1 \Sigma^3 \Omega = n \operatorname{tr}_1 \Sigma^3 + 3 \operatorname{tr}_1 \Sigma^3 \Omega$$

Differentiation of $X_3(\beta)$ yields:

$$\frac{d}{d\beta} (X_3(\beta)) = 6 X_4(\beta)$$

$$\text{where } X_4 = X_4(\beta)|_{\beta=0}$$

$$\begin{aligned} &= n \operatorname{tr}_1^4 \Sigma - 4n \operatorname{tr}_1^2 \Sigma \operatorname{tr}_2 \Sigma + 4n \operatorname{tr}_1 \Sigma \operatorname{tr}_3 \Sigma - 4n \operatorname{tr}_4 \Sigma + 2n \operatorname{tr}_2^2 \Sigma + 4 \operatorname{tr}_1 \Sigma^4 \Omega \\ &= n \operatorname{tr}_1 \Sigma^4 + 4 \operatorname{tr}_1 \Sigma^4 \Omega \end{aligned}$$

so that:

$$3.9 \quad E(\operatorname{tr}^4 A) = X_1^4 + 12 X_1^2 X_2 + 32 X_1 X_3 + 12 X_2^2 + 48 X_4$$

Similarly it follows that:

$$3.10 \quad E(\operatorname{tr}^5 A) = X_1^5 + 20 X_1^3 X_2 + 80 X_1^2 X_3 + 240 X_1 X_4 + 60 X_1 X_2^2 + 160 X_2 X_3 + 384 X_5$$

$$\text{where } X_5 = n \operatorname{tr}_1 \Sigma^5 + 5 \operatorname{tr}_1 \Sigma^5 \Omega$$

and

$$\begin{aligned} 3.11 \quad E(\operatorname{tr}^6 A) = & X_1^6 + 30 X_1^4 X_2 + 160 X_1^3 X_3 + 720 X_1^2 X_4 + 180 X_1^2 X_2^2 + 960 \\ & X_1 X_2 X_3 + 2304 X_1 X_5 + 1440 X_2 X_4 + 120 X_2^3 + 640 X_3^2 + 3840 X_6 \end{aligned}$$

where $X_6 = n \operatorname{tr}_1 \Sigma^6 + 6 \operatorname{tr}_1 \Sigma^6 \Omega$

The correctness of the above moments can be checked by substituting $\Omega = 0$ in equations 3.6, 3.7, 3.8 and 3.9. These give the same results as derived by Khatri and Pillai (1967) for the central Wishart distribution.

Discussion:

- (a) Using this technique, every moment of $\operatorname{tr} A$ can be calculated from the preceding factor X_i .

It will be noticed that the h -th moment ($h = 1, \dots, 6$) is the sum of products of the X_i 's. The factors X_i in each product form a partition of h , e.g. in 3.9 the partitions are:

$$(1^4), (1^2, 2), (1, 3), (2^2), (4) \text{ for } h = 4.$$

Conjecture:

If H is considered as a partition of the integer h , the h -th moment of $\operatorname{tr} A$, where $A \sim W'(\Sigma, n, \Omega)$, is given by:

$$3.12 \quad E(\operatorname{tr}^h A) = \Sigma_H \alpha_H X_H, \text{ where } H = (h_1, \dots, h_s) \text{ is a partition of } h \text{ and}$$

$$3.13 \quad X_k = (n \operatorname{tr}_1 \Sigma^k + k \operatorname{tr}_1 (\Sigma^k \Omega))$$

- (b) The following four moments are also of special interest:

$$3.14 \quad \mu'_1(\operatorname{tr} A) = X_1$$

$$3.15 \quad \mu_2(\operatorname{tr} A) = 2X_2$$

$$3.16 \quad \mu_3(\operatorname{tr} A) = 8X_3$$

$$3.17 \quad \mu_4(\operatorname{tr} A) = 12X_2^2 + 48X_4$$

4. The Moments of $\operatorname{tr} A_{ii}$, where A_{ii} is a submatrix of A

To study the moments of the submatrices A_{ii} of A , it is sufficient to partition A as:

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

where A_{ii} ($i = 1, 2, 3$) is a $p_i \times p_i$ submatrix of A and $p_1 + p_2 + p_3 = p$.

The moments of $\text{tr} A_{22}$ will be representative of the moments of $\text{tr} A_{ii}$.

Let

$$B(p \times p) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{p_2} & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

be partitioned similarly as A .

Then

$$B'AB = \begin{bmatrix} 0 & 0 & 0 \\ 0 & A_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

which will be denoted by A_{22}^* .

The moment generating function of A_{22} will be given by:

$$\begin{aligned} 4.1 \quad M_{A_{22}}(\theta_{22}) &= E(\text{etr} \theta_{22} A_{22}) = E(\text{etr}(AB\theta B')) = M_A(B'\theta B) \\ &= |I - 2B'\theta B\Sigma|^{-n/2} \text{etr}(\Omega B'\theta B\Sigma(I - 2B'\theta B\Sigma)^{-1}) \\ &= |I - 2\theta_{22}\Sigma_{22}|^{-n/2} \exp\{\text{tr}(\Omega_{12}\theta_{22}\Sigma_{21} \\ &\quad + 2\Omega_{12}\theta_{22}\Sigma_{22}(I - 2\theta_{22}\Sigma_{22})^{-1}\theta_{22}\Sigma_{21}) \\ &\quad + \text{tr}\Omega_{22}\theta_{22}\Sigma_{22}(I - 2\theta_{22}\Sigma_{22})^{-1} \\ &\quad + \text{tr}(2\Omega_{32}\theta_{22}\Sigma_{22}(I - 2\theta_{22}\Sigma_{22})^{-1}\theta_{22}\Sigma_{23} + \Omega_{32}\theta_{22}\Sigma_{23})\} \end{aligned}$$

when using the fact that if

$$C = \begin{bmatrix} C_{11} & 0 & 0 \\ C_{21} & C_{22} & C_{23} \\ 0 & 0 & C_{33} \end{bmatrix}$$

then:

$$|C| = |C_{22}|$$

and:

$$C^{-1} = \begin{bmatrix} C_{11}^{-1} & 0 & 0 \\ -C_{22}^{-1}C_{21}C_{11}^{-1} & C_{22}^{-1} & -C_{22}^{-1}C_{23}C_{33}^{-1} \\ 0 & 0 & C_{33}^{-1} \end{bmatrix}$$

Replacing θ_{22} in the moment generating function 4.1 by a scalar β , we arrive at the moment generating function of $\text{tr}A_{22}$ i.e.

$$\begin{aligned} 4.2 \quad M_{\text{tr}A_{22}}(\beta) &= |I - 2\beta\Sigma_{22}|^{-n/2} \cdot \exp B \{ \text{tr}(\Omega_{12}\Sigma_{21} + 2\beta\Omega_{12}\Sigma_{22}(I - 2\beta\Sigma_{22})^{-1}\Sigma_{21}) \\ &\quad + \text{tr}(\Omega_{22}\Sigma_{22}(I - 2\beta\Sigma_{22})^{-1}) \\ &\quad + \text{tr}(2\beta\Omega_{32}\Sigma_{22}(I - 2\beta\Sigma_{22})^{-1}\Sigma_{23} + \Omega_{32}\Sigma_{23}) \} \end{aligned}$$

Using Lemma 2 and 3 and Corollary 2 when differentiating to β , and then equating β to zero it follows that:

$$\begin{aligned} 4.3 \quad E(\text{tr}A_{22}) &= n \text{tr}\Sigma_{22} + \text{tr}(\Sigma_{21}\Omega_{12} + \Sigma_{22}\Omega_{22} + \Sigma_{23}\Omega_{32}) \\ &= n \text{tr}\Sigma_{22} + \text{tr}(\Sigma\Omega)_{22} \end{aligned}$$

Denoting this by X_1 and differentiating $X_1(\beta)$ successively to β , and equating β to zero it follows that:

$$\frac{d}{d\beta} X_1(\beta) = 2X_2(\beta)$$

$$\frac{d}{d\beta} X_2(\beta) = 4X_3(\beta)$$

$$\frac{d}{d\beta} X_3(\beta) = 6X_4(\beta) \quad \text{etc.}$$

and:

$$4.4 \quad E(\text{tr}^2 A_{22}) = X_1^2 + 2X_2$$

$$4.5 \quad E(\text{tr}^3 A_{22}) = X_1^3 + 6X_1X_2 + 8X_3$$

$$4.6 \quad E(\text{tr}^4 A_{22}) = X_1^4 + 12X_1^2X_2 + 32X_1X_3 + 12X_2^2 + 48X_4$$

where:

$$X_2 = n \text{tr}_1^2 \Sigma_{22} - 2n \text{tr}_2 \Sigma_{22} + 2 \text{tr}_1 (\Sigma_{22} \Sigma_{21} \Omega_{12} + \Sigma_{22}^2 \Omega_{22} + \Sigma_{22} \Sigma_{23} \Omega_{32})$$

$$X_3 = n \text{tr}_1^3 \Sigma_{22} - 3n \text{tr}_1 \Sigma_{22} \text{tr}_2 \Sigma_{22} + 3n \text{tr}_3 \Sigma_{22} +$$

$$+ 3 \text{tr}_1 (\Sigma_{22}^2 \Sigma_{21} \Omega_{12} + \Sigma_{22}^3 \Omega_{22} + \Sigma_{22}^2 \Sigma_{23} \Omega_{32})$$

$$X_4 = n \text{tr}_1^4 \Sigma_{22} - 4n \text{tr}_1^2 \Sigma_{22} \text{tr}_2 \Sigma_{22} + 4n \text{tr}_1 \Sigma_{22} \text{tr}_3 \Sigma_{22} - 4n \text{tr}_4 \Sigma_{22}$$

$$+ 2n \text{tr}_2^2 \Sigma_{22} + 4 \text{tr}_1 (\Sigma_{22}^3 \Sigma_{21} \Omega_{12} + \Sigma_{22}^4 \Omega_{22} + \Sigma_{22}^3 \Sigma_{23} \Omega_{32})$$

Thus the first four moments of $\text{tr} A_{ii}$ are given by expressions similar to 4.3, 4.4, 4.5 and 4.6 where:

$$4.7 \quad X_k = n \text{tr}_1 \Sigma_{ii}^k + k \text{tr}_1 (\Sigma_{ii}^{*k} \Sigma \Omega)_{ii}, \quad k = 1, \dots, 4$$

In general it thus seems that 3.13 becomes

$$4.8 \quad X_k = n \text{tr}_1 \Sigma_{ii}^k + k \text{tr}_1 (\Sigma_{ii}^{*k} \Sigma \Omega)_{ii}$$

when submatrices of A and Σ are considered.

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